Blender Workflow and Limitations

Overview

Blender was explored during the early stage of the project to improve the visual realism of the lung model and to understand how different disease stages could be visually represented. The intention was to create separate models for healthy, early-stage, and advanced disease lungs, and evaluate whether these could be used in the main system pipeline.

Step 1: Importing and Preparing the Lung Model

The process started by importing the lung model into Blender as a `.glb` file. Once imported, the model was inspected in the viewport to ensure the geometry was intact and usable. Basic adjustments such as positioning and viewing were done to make the model easier to work with.

At this stage, the focus was on making sure the lung model could act as a consistent base for further modifications, rather than changing its geometry directly.

Step 2: Creating Multiple Disease Stages

After preparing the base model, it was duplicated multiple times to represent different stages of disease progression. These duplicates were intended to act as independent versions of the lung for:

- healthy stage
- early disease stage
- advanced disease stage

Each copy was then modified separately using different material and texture settings. The idea was to visually represent how lung tissue changes as disease progresses.

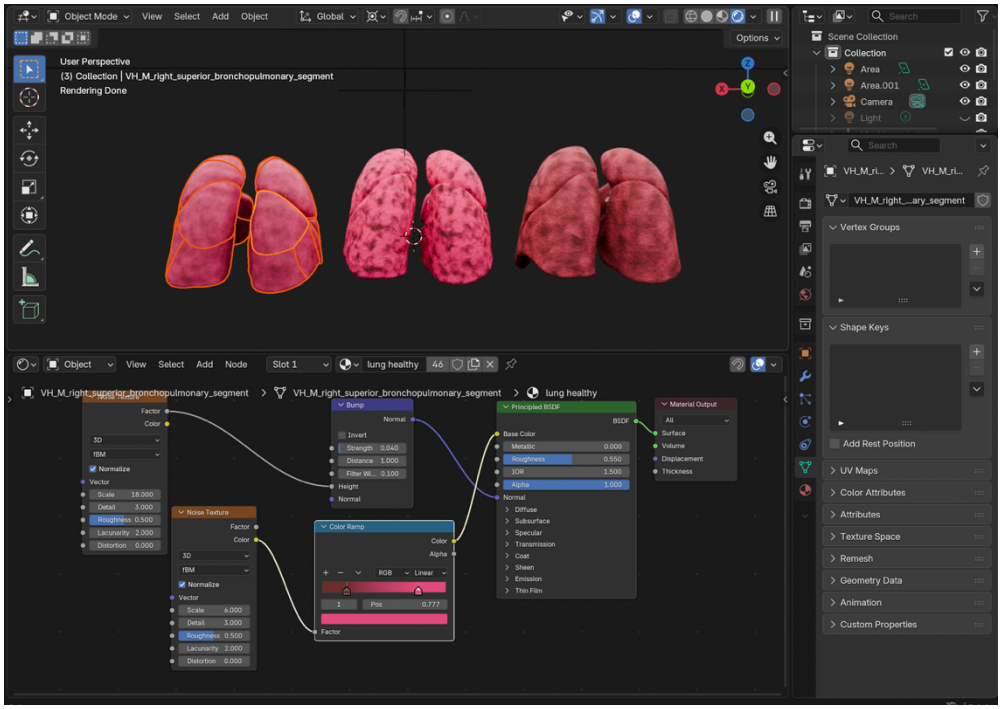

Figure 1: Multiple lung stage versions (healthy, early, advanced) created by duplicating the base model.

Step 3: Designing Materials to Simulate Disease

To create realistic disease patterns, Blender's shader node system was used. The main approach involved combining procedural textures and colour control to simulate irregular lung abnormalities.

Noise Texture nodes were used to generate random, uneven patterns across the lung surface. These patterns were passed through a ColorRamp node to control how colours were distributed, allowing certain areas to appear darker or more affected.

To further improve realism, a Bump node was introduced. This added small variations to the surface normals, making the lung appear slightly rough instead of perfectly smooth. All of these components were connected to the Principled BSDF shader, which controlled the final material appearance including colour, roughness, and lighting response.

For the healthy lung, the material was kept simple, with a smooth surface and light pink colour. For the early stage, mild irregularities and slightly darker regions were added. For the advanced stage, stronger texture variation, darker tones, and more noticeable surface roughness were applied.

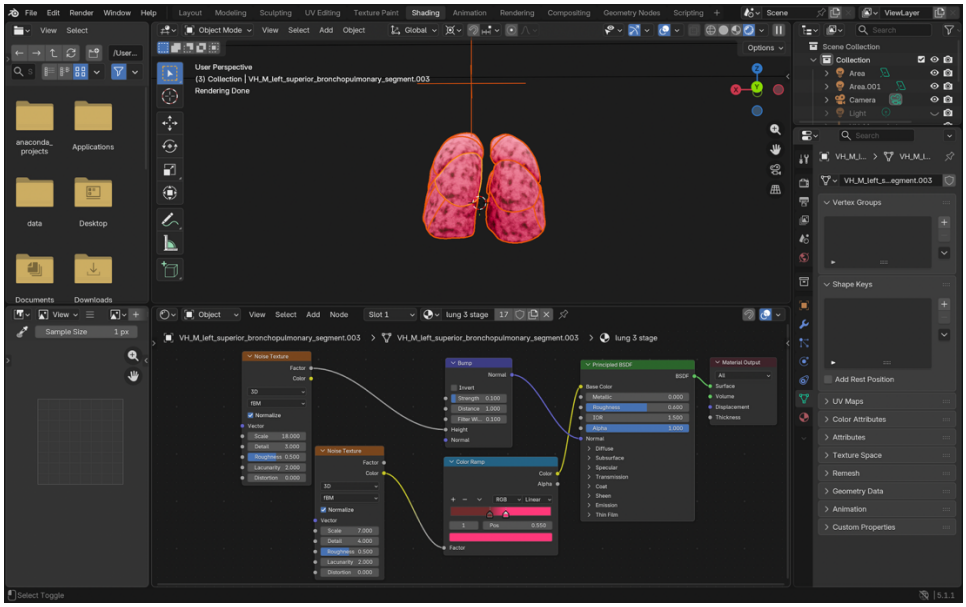

Figure 2: Shader node setup used to generate disease texture patterns using Noise Texture and ColorRamp.

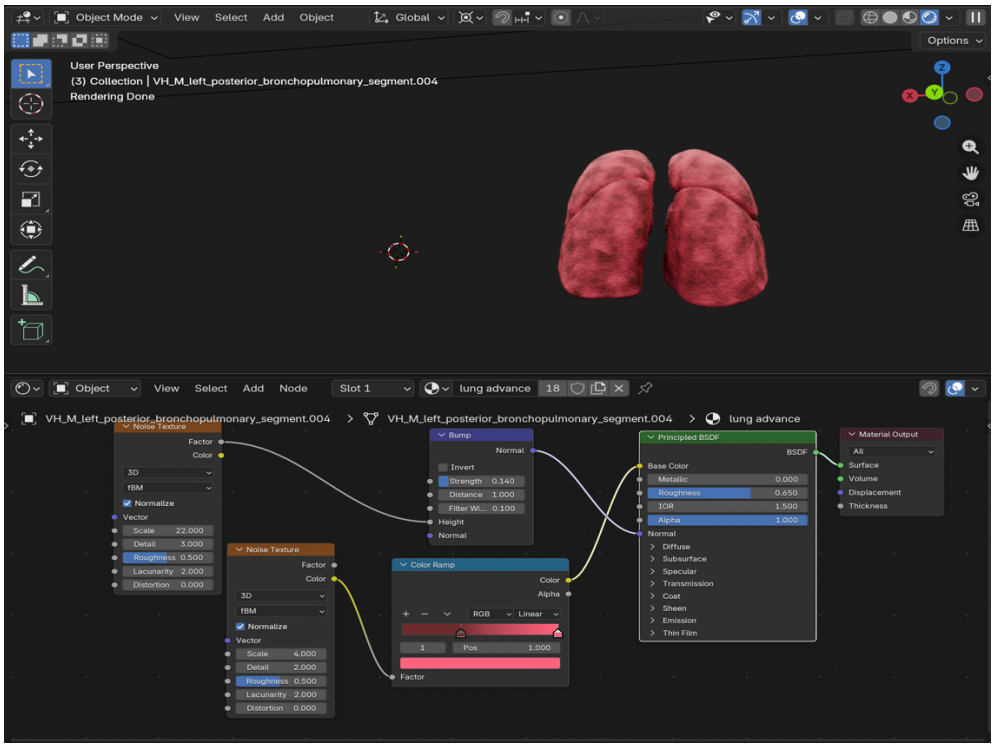

Figure 3: Advanced stage material showing stronger texture, darker colours, and increased surface roughness.

Step 4: Lighting Setup for Realism

After creating the materials, lighting was adjusted to improve how the model appeared in the scene. Multiple light sources were tested, including area lights positioned around the model.

The goal of this step was to reduce the flat appearance of the lung and highlight surface details created by the material. By adjusting light intensity, angle, and placement, the model appeared more realistic and closer to medical-style visual references.

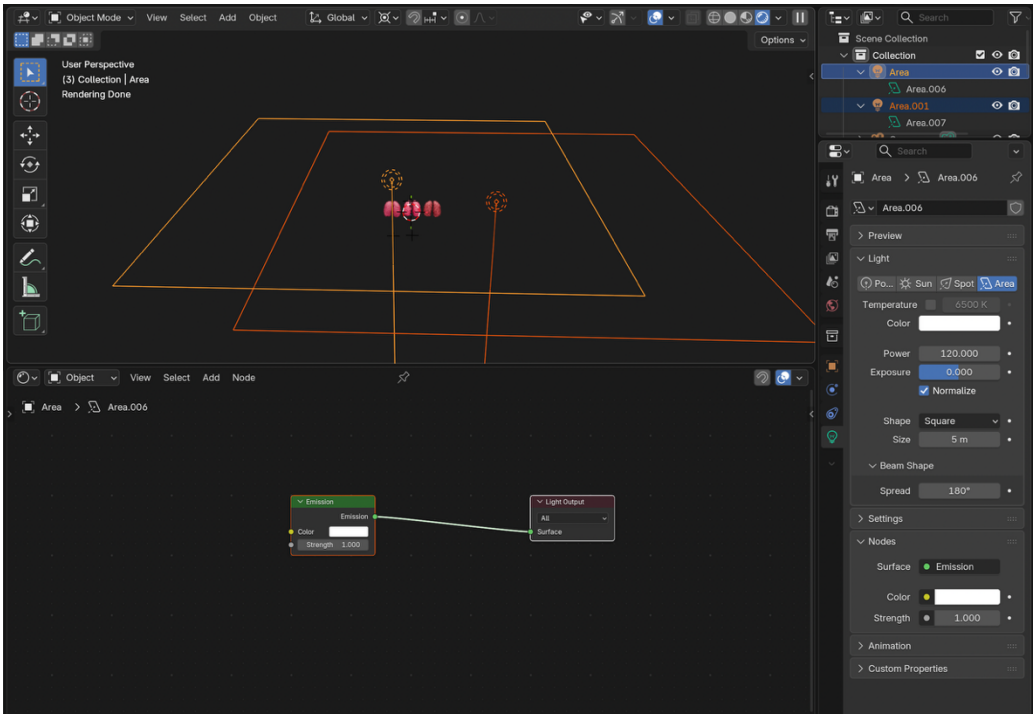

Figure 4: Lighting setup using area lights to improve rendering quality and surface visibility.

Step 5: Observed Results

Through this workflow, the Blender models showed clear visual differences between healthy, early, and advanced stages. The surface texture, colour variation, and lighting all contributed to making the disease progression more visually understandable.

This stage helped the team understand how disease patterns could be represented visually, especially in terms of surface irregularity and colour change.

Step 6: Limitations Encountered

Despite producing good visual results, several practical issues were encountered.

One major problem was that duplicated lung models shared linked materials. This meant that when a material was modified for one stage, it sometimes affected the other stages as well, making it difficult to maintain independent representations.

Another issue occurred during export. The materials created using procedural nodes such as Noise Texture, ColorRamp, and Bump were not reliably preserved when exporting the model to .glb. As a result, the exported model often did not match what was seen inside Blender.

Additionally, Blender is mainly designed for static rendering, whereas the project required dynamic progression, where disease severity could change interactively. This type of parameter-driven control was difficult to achieve within Blender.

Step 7: Final Decision

Due to these limitations, Blender was not used in the final implementation. Instead, it was used as a visual experimentation tool to understand how disease stages could be represented.

The project was then moved to a Python-based approach using PyVista, which allowed better control over disease progression, deformation, and interaction through parameters such as sliders.

Conclusion

Blender played an important role in exploring visual realism and understanding how disease progression could look. However, due to limitations in export reliability and dynamic control, it was not suitable for the final pipeline. The insights gained from this stage were used to improve the final implementation in Python.